GDAŃSK UNIVERSITY OF TECHNOLOGY

Documentation of Completed Tasks

2025/2026

**High Availability Distributed Systems**

*Task 1: IoT - Cisco*

Field of Study: Computer Science

Team Members:
*Mateusz Fydrych s193410*
*Adam Chabraszewski s193373*

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# 1 Introduction

## 1.1 Roles in the implementation of tasks

During the completion of the assigned tasks, no fixed roles were established within the team. We worked collaboratively on all parts of the project in order to develop a broader understanding of the discussed issues and the practical aspects of the course. Although this approach was not the most time-efficient, it proved highly effective in enabling us to achieve a deeper and more comprehensive understanding of the subject matter.

## 1.2 The purpose of the task

The purpose of this task was to become familiar with a new software environment, namely *Cisco Packet Tracer*, and to develop a fundamental understanding of High Availability Distributed Systems, Computer Networks, and, in particular, the Internet of Things. For both team members, this was the first experience with this software. In this report, we present the key milestones and the results of our work related to the completion of the course entitled *Exploring Internet of Things with Cisco Packet Tracer*. We completed all theoretical modules together with the corresponding assessments and followed each practical exercise step by step. This approach allowed us to achieve the intended learning outcomes in a thorough and structured manner.

# 2 Assumptions for the implementation of the task

## 2.1 Technical and non-technical assumptions

The task was carried out using *Cisco Packet Tracer* and the materials provided in the course. It was assumed that both team members had access to a computer with the software installed and were able to complete both the theoretical and practical parts of the module. In practice, the software was used on Windows, as macOS support is limited in this context.

Since this was our first contact with *Cisco Packet Tracer* in the context of IoT, we also assumed that the implementation would require a gradual, step-by-step approach. Another important assumption was that the task would be completed collaboratively, allowing us to better understand the presented concepts and practical activities.

## 2.2 Technology stack

The task was completed using the following tools and technologies:

- *Cisco Packet Tracer*,
- IoT devices available in the simulator,
- course materials, including theoretical content, videos, and Packet Tracer activities,
- *Overleaf*

## 2.3 Expected results of the task implementation

The report includes screenshots accompanied by descriptions of the completed course content. The presentation of each module follows a consistent structure. First, a theoretical overview summarizes the knowledge acquired in a given module. Next, the main milestones of that module are presented together with relevant screenshots. The conclusion section summarizes the key takeaways from the course in a concise form. An additional section also includes supplementary information, such as extra screenshots that were considered less essential for inclusion in the main body of the report.

# 3 Implementation of the task

## 3.1 Module 1

Module 1 served as an introduction to the course and to the *Cisco Packet Tracer* environment in the context of the Internet of Things. It combined theoretical content, instructional videos, and practical simulation tasks. The module introduced the concept of smart home networks and presented the basic methods of adding, connecting, monitoring, and controlling IoT devices in a simulated environment.

The structure of the module allowed us to progress gradually from introductory material to more advanced practical activities. First, we became familiar with the purpose and scope of the course. Next, we explored the IoT devices available in Packet Tracer and learned how to integrate them into a smart home network. Finally, we completed activities related to connecting devices through a home gateway and controlling them with the use of a registration server.

### 3.1.1 Theoretical overview

Module 1 introduced the fundamental concepts of the Internet of Things in the *Cisco Packet Tracer* environment. It began with general course information and an explanation of the importance of IoT systems in modern networking. This part helped establish the context for the practical activities completed later in the module.

The module then presented IoT devices available in Packet Tracer and explained how they can be used in a smart home environment. It covered both wired and wireless devices, as well as the basic principles of adding them to an existing network. This provided the theoretical foundation for the first practical task involving the expansion of a smart home infrastructure.

Another topic was the role of the home gateway in connecting and managing devices within a local network. The module also introduced monitoring functions and the use of end-user devices such as laptops and smartphones. In addition, Bluetooth communication was presented as an example of short-range connectivity in IoT systems.

The final part of the module focused on registration servers and centralized device control. It explained how IoT devices can be registered, managed remotely, and accessed through additional user devices.

### 3.1.2 Practical tasks

The first milestone was the completion of task 1.0, which served as an introductory subsection. At this stage, no advanced technical knowledge was required.

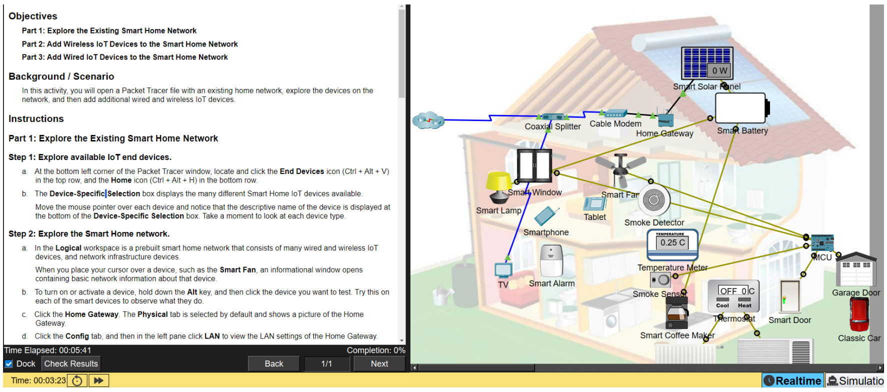


Figure 1: Initial topology of task 1.1.3 – Packet Tracer: Add IoT Devices in PT.

This was our first encounter with the application. We observed that the project environment was well structured. On the left side, the tasks to be completed were clearly visible, while on the right side the logical architecture was already prepared for further extension. In the lower-left corner, the interface provided convenient access to adding new devices and connections.

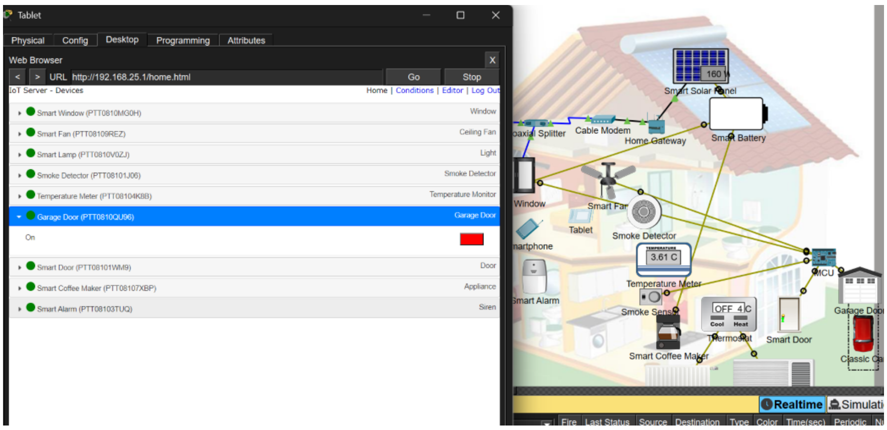


Figure 2: Interactive task 1.1.3 – Interactive IoT – Door

After logging in to the home gateway (`192.168.25.1` – the same IP address was also used in other tasks), we had an opportunity to test an interactive activity involving the opening and closing of a garage door.

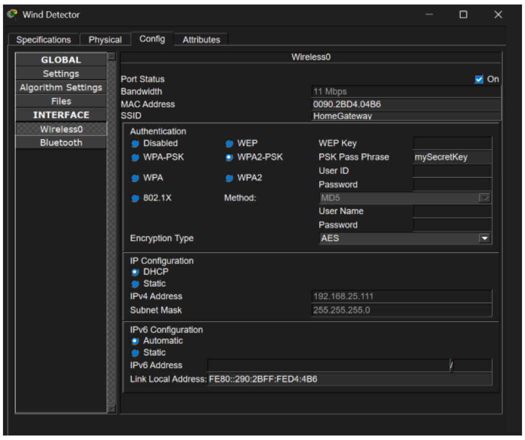


Figure 3: Final configuration of the Wind Detector device

The figure presents the final configuration of the Wind Detector device, which enabled the successful completion of the task.

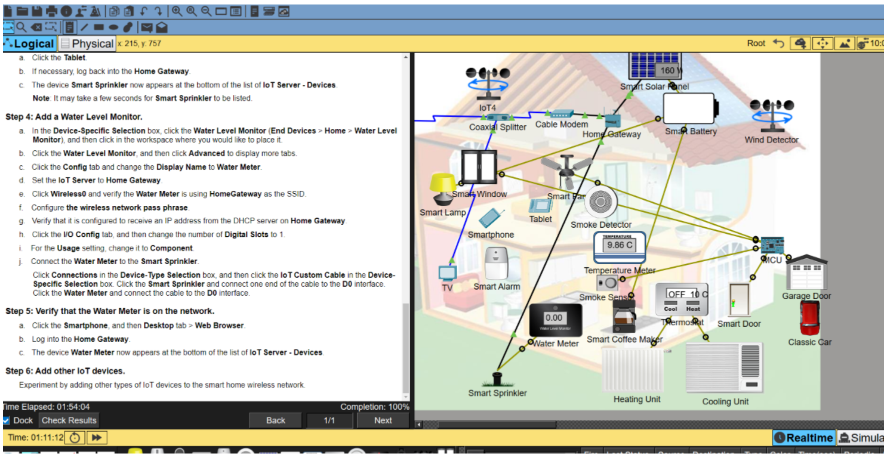


Figure 4: Final result of task 1.1.3 – Packet Tracer: Add IoT Devices in PT.

The figure above presents the network created during the first practical task. As this was our initial hands-on activity, it took us 1 hour and 54 minutes to complete. The application occasionally stopped responding, and several restarts were required. Nevertheless, completing the final step and seeing the Completion 100% message was highly satisfying.

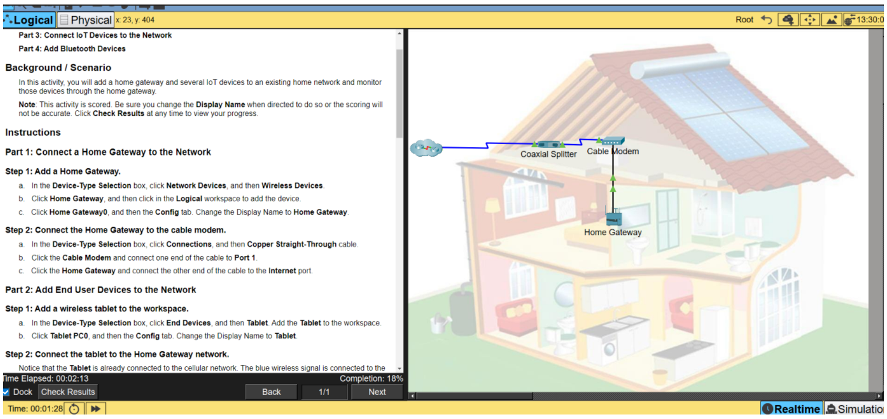


Figure 5: Successfully completed part 1 of task 1.2.3 – the home gateway added to our network

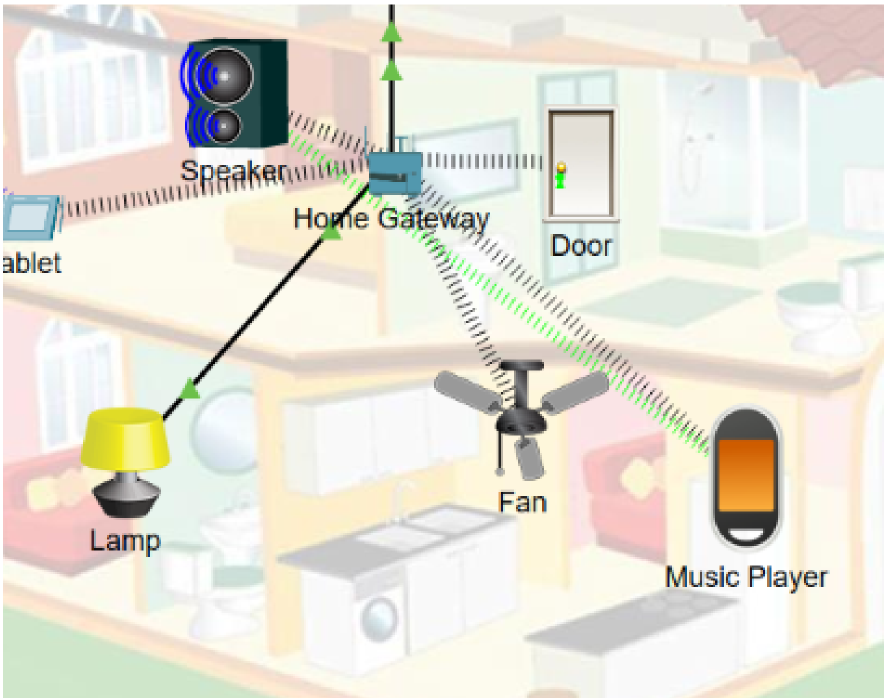


Figure 6: The speaker successfully playing music from the music player device – final stage of the task

It was interesting to observe that Cisco Packet Tracer allows the simulation of realistic multimedia interactions, such as playing music. The Bluetooth connection was established successfully, which enabled audio playback between devices.

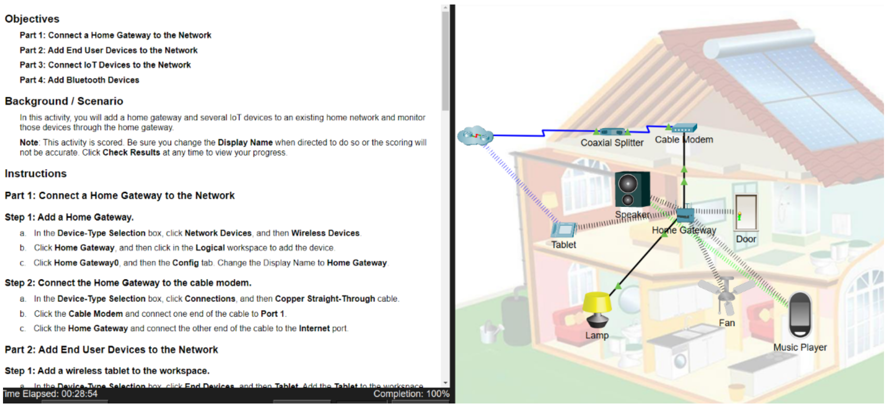


Figure 7: Final result of task 1.2.3 – Packet Tracer: Connect Devices to a Home Gateway and Monitor Network.

This figure presents the final architecture of the network. Although it is less complex than the previous one, this task introduced Bluetooth connectivity in its final stage, which allowed us to play music remotely. This exercise expanded our understanding of communication methods used in IoT environments.

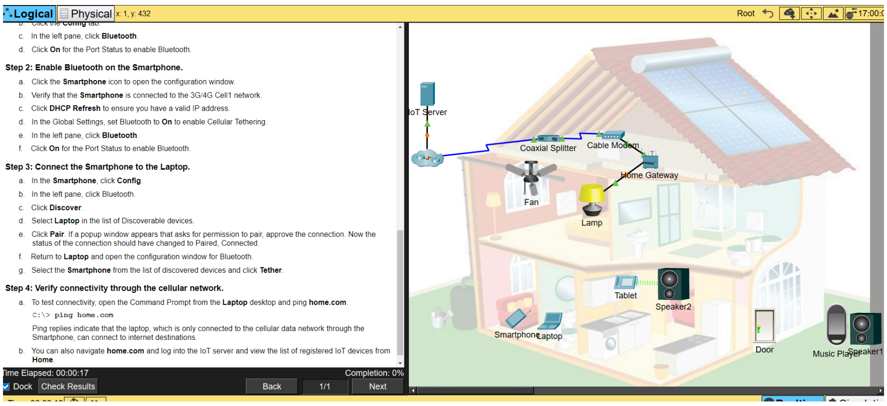


Figure 8: Initial state of task 1.2.6 – Packet Tracer: Connect and Control Devices Using a Registration Server.

At the beginning of this task, we could observe a realistic example of a basic logical topology in a typical smart home environment.

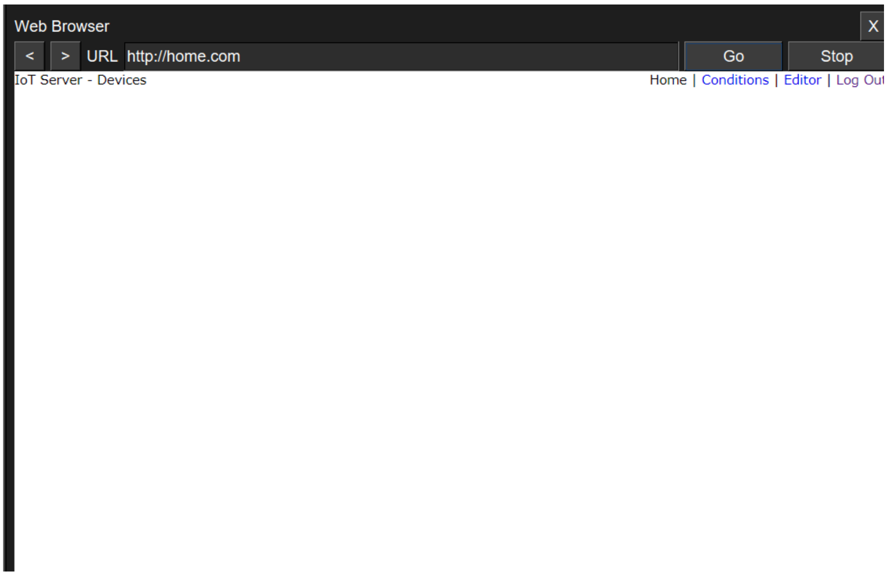


Figure 9: Expected behavior in task 1.2.6 – no IoT devices had been added yet

No IoT devices had been added to the remote server at this stage, so, as expected, no devices were listed.

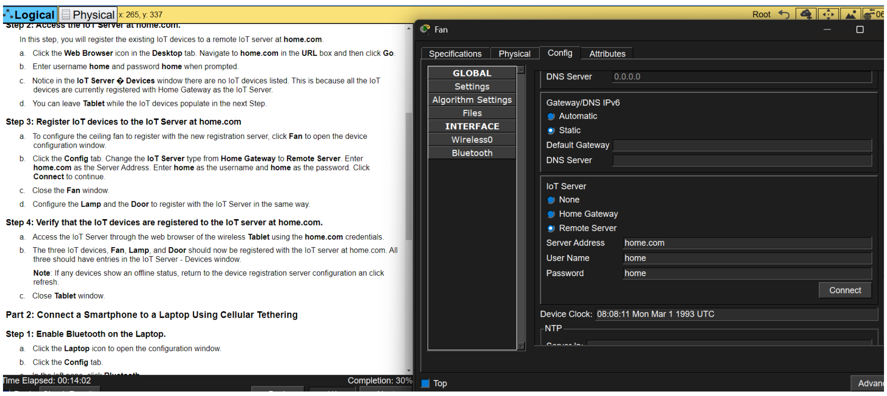


Figure 10: Completed Step 3 in task 1.2.6 – device credentials configured

Following the instructions, the fan was assigned the username and password: “home”. However, it is important to note that this is not a good security practice. First, using identical values for the username and password makes authentication easier to guess. Second, the password is very short. Finally, using the same value as both the credentials and the domain name may increase vulnerability to simple dictionary-based attacks.

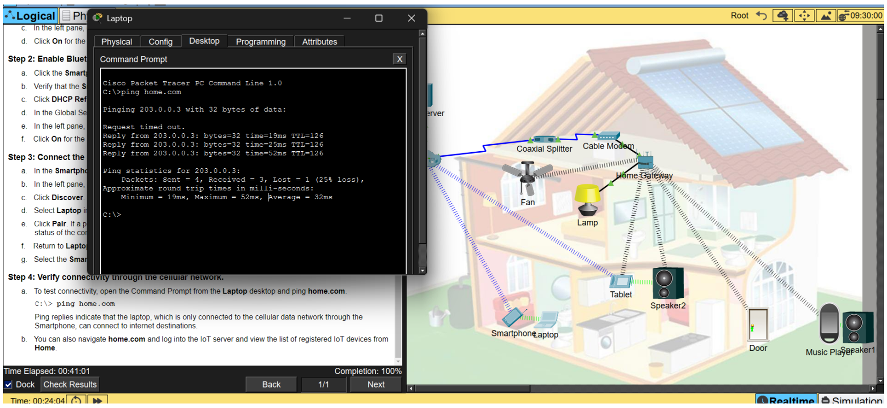

Figure 11: Final result of task 1.2.6 – Packet Tracer: Connect and Control Devices Using a Registration Server.

In the screenshot, we can observe that `ping home.com` returns a successful response, and the completion status is 100%. This task focused on connecting and managing IoT devices through a registration server. It provided practical insight into remote device control and demonstrated how centralized services can be used to manage multiple smart devices within a networked environment. The key takeaway from this task was the discovery that IoT devices can be connected to remote servers for centralized management and control.

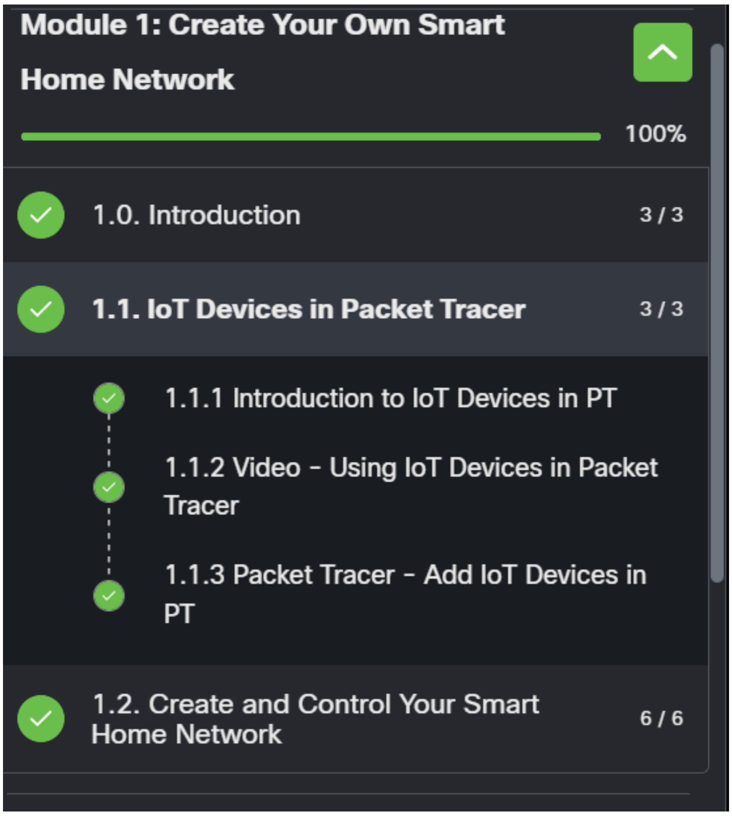


Figure 12: Module 1 completed successfully.

Module 1 was primarily focused on an introductory overview of network devices and the fundamentals of building basic computer networks. It served as an effective first step into the Cisco Packet Tracer environment and helped us understand the core concepts necessary for more advanced tasks in later modules.

## 3.2 Module 2

Module 2 focused on environmental controls and the creation of custom IoT Things in the *Cisco Packet Tracer* environment. In contrast to Module 1, which concentrated primarily on connecting and managing devices in a smart home network, this part of the course introduced a broader perspective by showing how environmental conditions can influence IoT systems and how programmable devices can be created to respond to such changes.

The module combined theoretical explanations, instructional videos, and practical Packet Tracer activities. It introduced the concept of environmental parameters in the Physical Workspace, explained how containers operate, and demonstrated how different environmental elements may affect the behavior of IoT devices. In the later part of the module, attention shifted to the design and modification of custom IoT Things, including the use of scripts and the programming capabilities available in Packet Tracer.

### 3.2.1 Theoretical overview

The first part of Module 2 introduced environmental controls available in *Cisco Packet Tracer*.


Figure 13: Example of environmental parameters in Cisco Packet Tracer

In the Physical Workspace, each container, such as the intercity area, has its own set of environmental values. These values include parameters such as temperature, rain, water level, and wind speed. The module demonstrated that IoT devices may either affect these environmental conditions or react to them.

The module also introduced several key terms necessary for interpreting environmental simulation in Packet Tracer:

**Important terms and concepts:**

- *Current time* – the simulated time within a container, which advances in 30-minute intervals. In this model, 1 second of real time corresponds to 30 minutes of container time. The timer ranges from 0:00 (midnight) to 11:59.

- *KeyFrame* – a representation of a single moment in time within the simulation.
- *KeyFrame graph* – a graph illustrating the value of environmental elements at specific points in time throughout the day.
- *Transference* – a set of values that determines the rate at which a child container converges with its parent container. This mechanism operates in the same manner for all environmental parameters.

The second major topic of Module 2 concerned the creation of custom IoT Things. The course explained that Packet Tracer allows users not only to work with predefined devices but also to create their own programmable objects. In addition, appropriate graphics can be selected to represent different states of a device. The recommended approach is to identify an existing Thing with similar functionality and then modify its script accordingly.

This part of the module also emphasized the importance of programming knowledge in the development of IoT solutions. Packet Tracer supports *JavaScript*, *Python*, and *Visual Blocky*, which makes it possible to adapt existing device behavior or create new logic from scratch.

### 3.2.2 Practical tasks

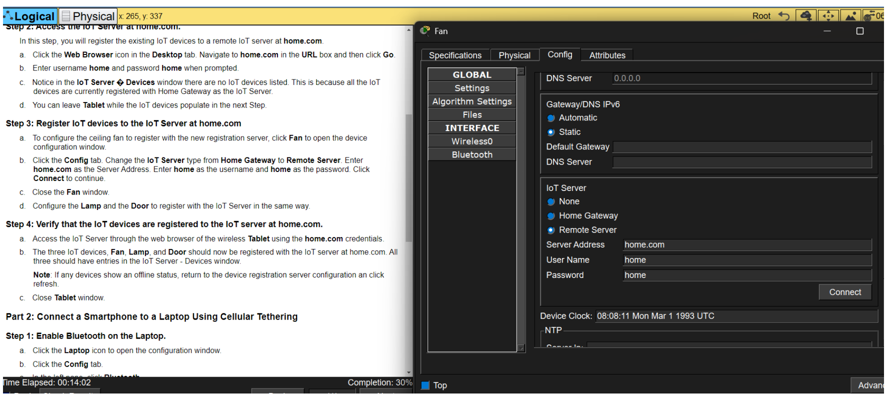


Figure 14: Task 2.0.5 – Initial project view

The first practical task was 2.0.5 – Packet Tracer - Modify and Monitor Environmental Controls in Packet Tracer.

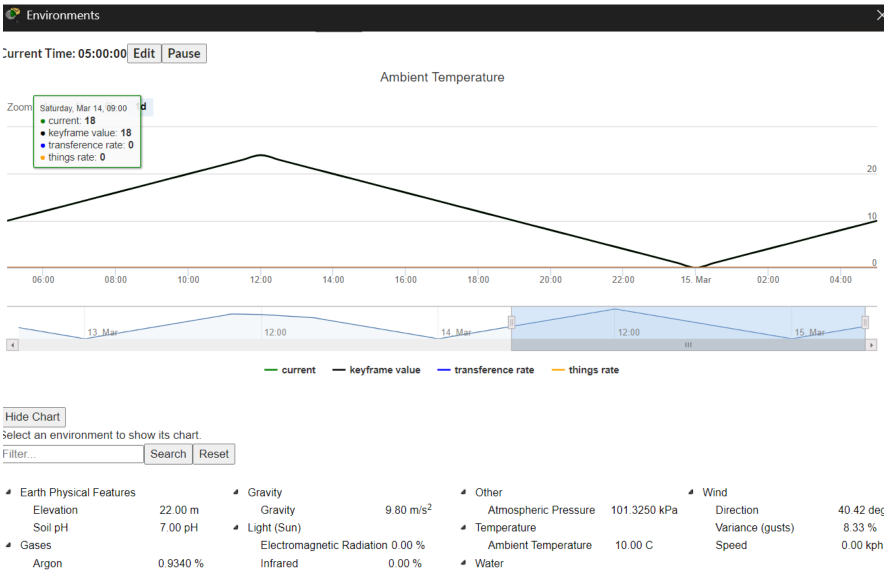


Figure 15: Task 2.0.5 – Ambient Temperature view

During this step, we observed that adjusting environmental parameters and time in Cisco Packet Tracer is relatively straightforward. This feature is particularly useful in a sandbox environment. In addition, the level of detail available for troubleshooting problems and simulating realistic scenarios is significant.

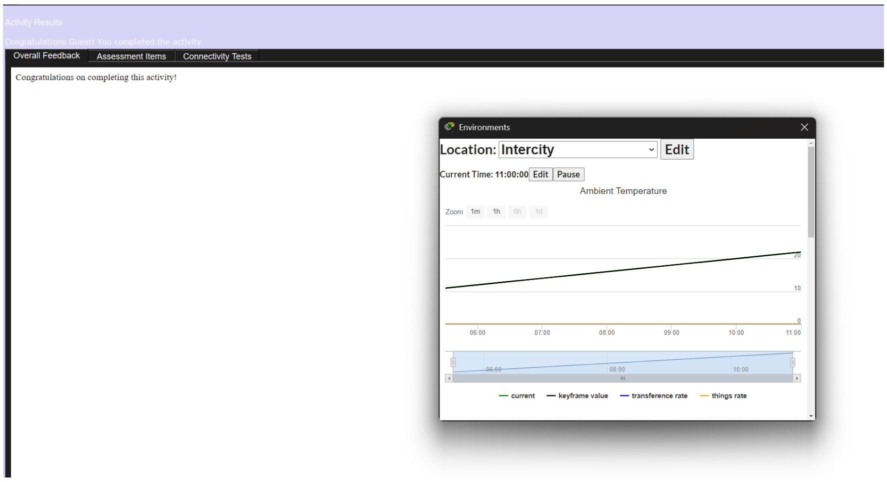


Figure 16: Task 2.0.5 – Final result

The second practical task was 2.1.3 – Packet Tracer - Create Your Own IoT Thing.

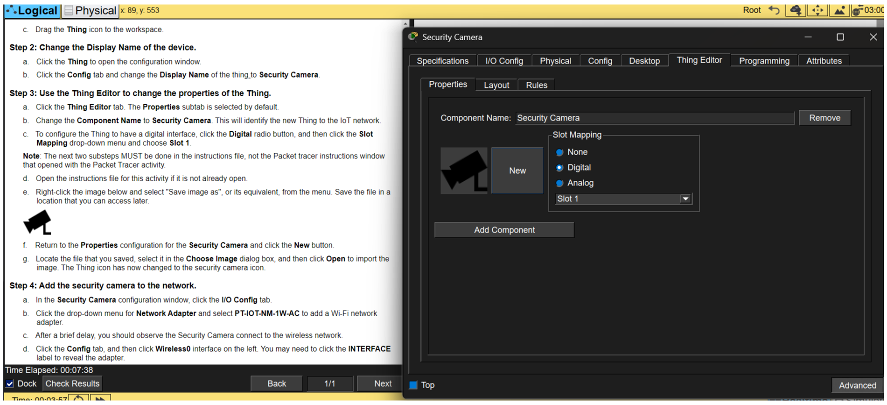


Figure 17: Task 2.1.3 – Successfully added new IoT device – Security Camera

After saving the camera graphic from the online Cisco course as a PNG file, we successfully added a new icon.

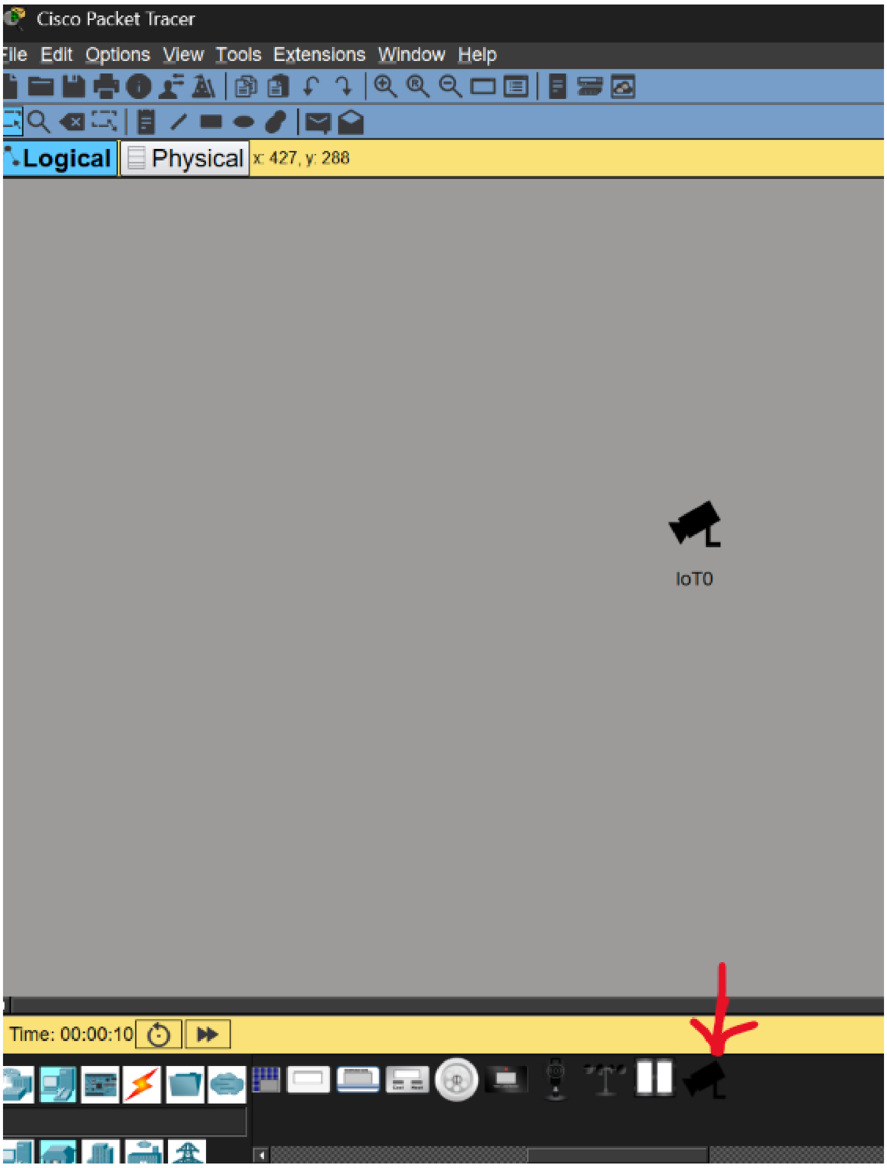


Figure 18: Task 2.1.3 – Final result – security camera added as a new IoT template

The final practical task was 2.1.6 – Modify an Existing Script for an IoT Thing.


Figure 19: Task 2.1.6 – Final result – Security Camera

As the final result, after modifying the `main.js` file, we added functionality to the Security Camera that enables it to detect motion as expected. The camera turns on a red light to indicate that movement has been detected.

In this task, programming skills were particularly useful for experimenting with the available options. There are many possible configurations and parameters that can be adjusted. The previous submodule 2.1.5 provided an excellent introduction to this activity.

Module 2 introduced a more advanced perspective on IoT systems in *Cisco Packet Tracer* by focusing on environmental controls, the addition of new IoT devices, and programmable Things. It showed that IoT solutions do not operate in isolation, but rather interact continuously with the physical environment in which they are deployed. Overall, this module broadened our understanding of Packet Tracer as a simulation environment. We learned not only how to observe and modify environmental conditions, but also how to create and adapt IoT Things to perform specific tasks. Overall, Module 2 served as an important bridge between basic IoT configuration and more advanced system design.


Figure 20: Final Exam score

As the final step, we took the "Final Exam". Although it consisted of only five questions, it served as a valuable test of whether we had understood the material in depth. We scored 100%, which confirms our understanding of the topic.


Figure 21: The certificate

# 4 Conclusions

- *Observations*

  We observed that Cisco Packet Tracer is an excellent tool for designing local networks. It is far more cost-effective to test network architectures in a simulated environment than to build them physically and only later discover configuration errors. Cisco has also equipped the application with accessible troubleshooting tools. For example, a laptop can be used similarly to a real device through the command-line interface, which allows the execution of basic diagnostic commands such as `ping` and `tracert`. However, we also noticed that the application occasionally became unresponsive and required a restart. Despite this limitation, the overall usefulness of the program remains very high.

- *Achievements*

  We successfully completed all practical tasks, finished the course, and obtained the *Exploring IoT with Cisco Packet Tracer* certificate.

- *Development potential*

  Each practical task provides a ready-made environment that can be further expanded. A simple example is task 1.1.3, in which we added an additional Wind Detector as IoT4. Cisco Packet Tracer offers many other IoT devices that could be incorporated into a more realistic smart home topology.

# 5 Additional screens

This section presents additional screenshots from the practical tasks that were considered less essential for the main body of the report.

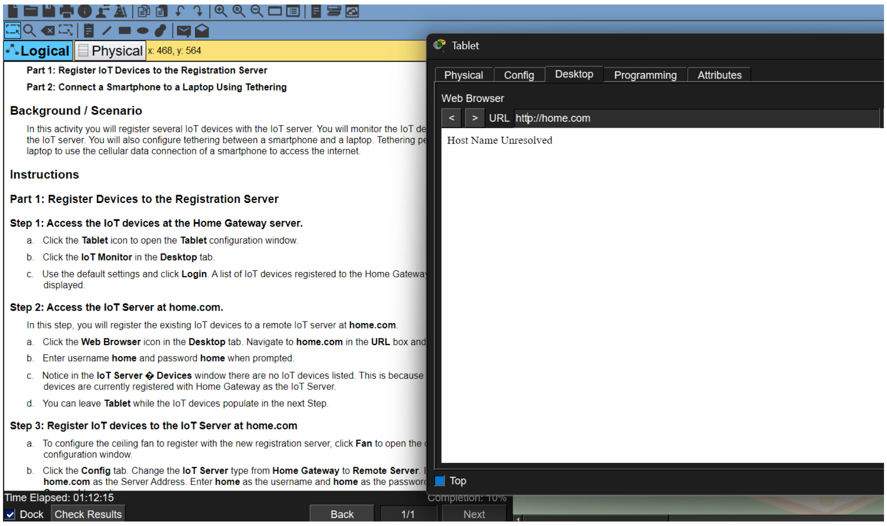

Figure 22: Task 1.1.3 – Example configuration – the figure presents the Wind Detector configuration


Figure 23: Task 1.2.6 – Example of successful Bluetooth pairing – smartphone connected to laptop

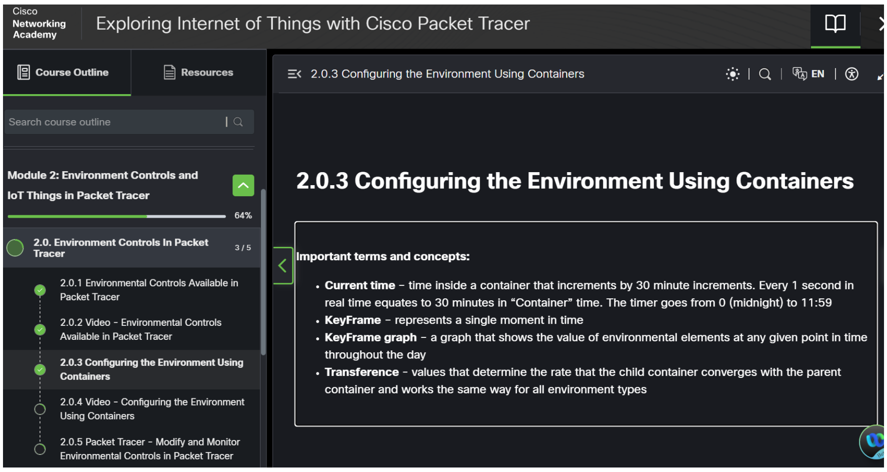


Figure 24: Module 2 – key definitions

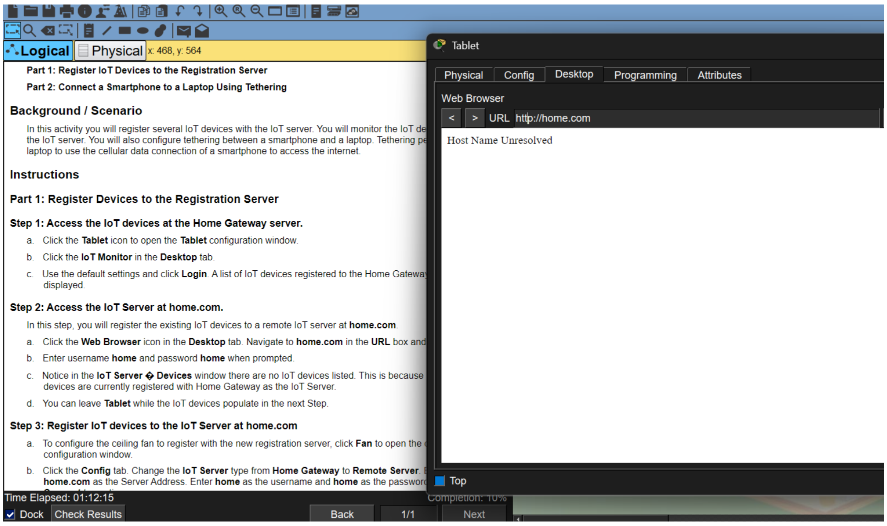


Figure 25: Example of a Cisco Packet Tracer application lag – after restarting the application, everything worked correctly.

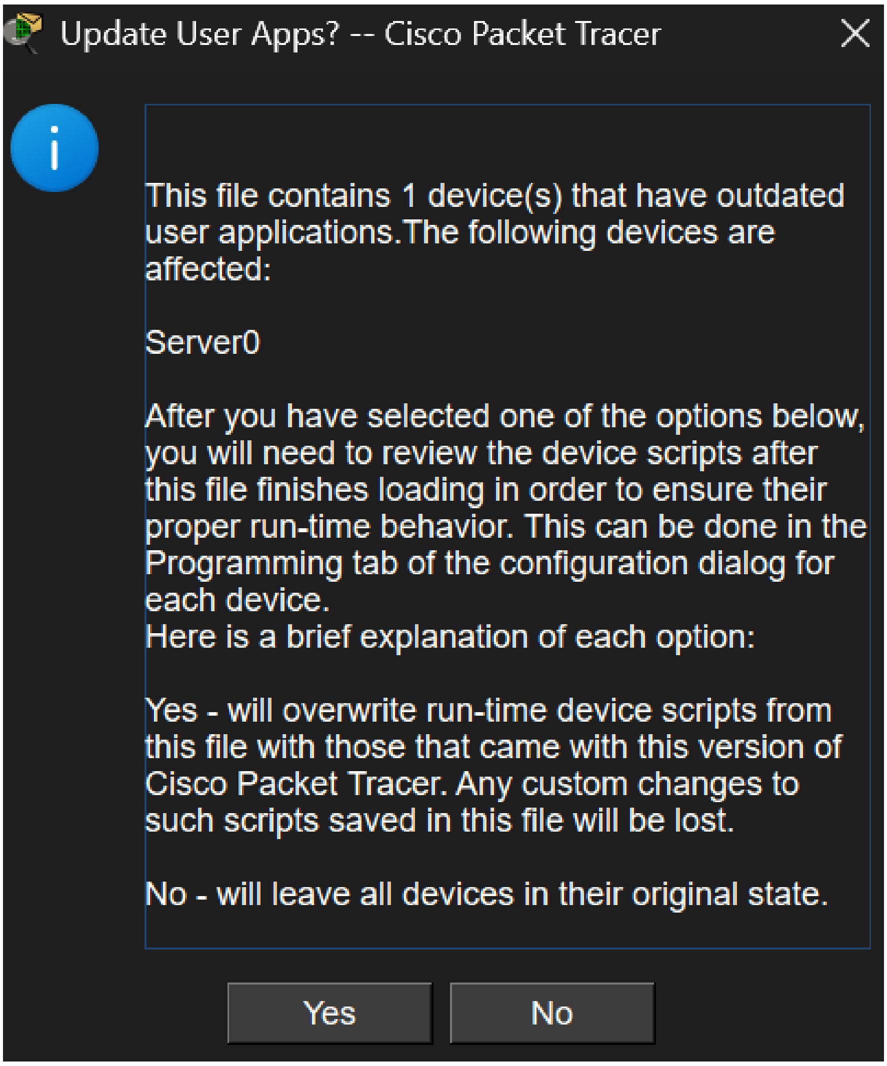


Figure 26: Example notification at the start of each task – usually the task was not updated to the latest version

All solutions are stored on our laptop and were completed manually by the authors.